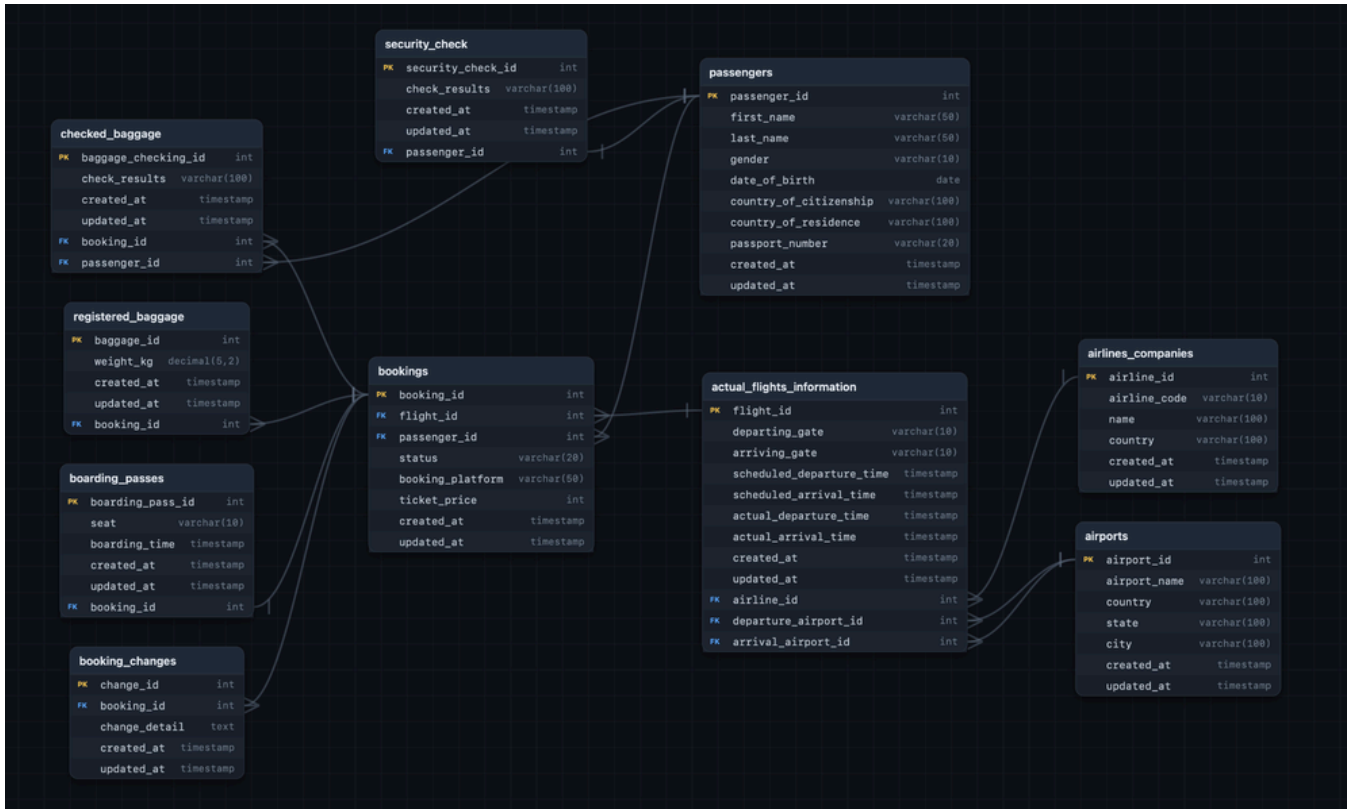


LABORATORY WORK 1

ERD Diagram: International Airport Database

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Entity-Relationship Diagram of the International Airport Database.

1.LEGEND

Symbol / notation	Meaning
PK	Primary Key -uniquely identifies a record in an entity
FK	Foreign Key - references a primary key in another entity
1:N	One-to-Many relationship
1:1	One-to-One relationship

2. ENTITY DESCRIPTIONS

Entity	Description
Airport	Contains airport details: ID, name, country, state, city, along with creation and modification timestamps.
Flight	Holds planned and actual flight data, gate assignments, airline ID, and departure/arrival airport IDs.
Airline	Maintains airline data, including ID, code, company name, country, and creation/update dates.
Booking	Stores passenger reservation records, including booking ID, flight ID, status, platform, timestamps, price, and passenger ID.
Boarding Pass	Holds boarding pass information, such as pass ID, seat number, boarding time, timestamps, and the related booking ID.
Baggage	Records details about registered luggage, including baggage ID, weight in kg, creation/update dates, and booking ID.
Baggage Checking	Stores luggage inspection outcomes, including inspection ID, status, timestamps, booking ID, and passenger ID.
Passenger	Holds passenger profile data, including ID, full name, gender, birth date, citizenship, residence, passport number, and timestamps.
Security Check	Stores airport security screening results, including check ID, status, timestamps, and passenger ID.
Booking Changes	Records modifications made to reservations, including change ID, booking ID, modification details, and timestamps.

3. RELATIONSHIP AND CARDINALITY

Relationship	Description
Airline - Flight	One-to-Many (1:N). Each flight is operated by an airline via <u>Flight.airline_id</u> .
Airport - Flight (Departure)	One-to-Many (1:N). A flight links to its departure airport via <u>Flight.departure_airport_id</u> .
Airport - Flight (Arrival)	One-to-Many (1:N). A flight links to its arrival airport via <u>Flight.arrival_airport_id</u> .
Flight - Booking	One-to-Many (1:N). Reservations reference a specific flight using <u>Booking.flight_id</u> .
Passenger - Booking	One-to-Many (1:N). Reservations are associated with passengers via <u>Booking.passenger_id</u> .
Booking - Boarding Pass	One-to-One (1:1). In this system model, a boarding pass is directly linked to a reservation via <u>booking_id</u> .
Booking - Baggage	One-to-Many (1:N). Registered luggage is tied to a reservation via <u>Baggage.booking_id</u> .
Booking - Baggage Checking	One-to-Many (1:N). Baggage inspection logs connect to a reservation using <u>Baggage_Checking.booking_id</u> .
Passenger - Baggage Checking	One-to-Many (1:N). Baggage inspection logs reference a passenger via <u>Baggage_Checking.passenger_id</u> .
Passenger - Security Check	One-to-Many (1:N). Security check records are connected to passengers via <u>Security_Check.passenger_id</u> .
Booking-Booking Changes	One-to-Many (1:N). A reservation can have several modification entries, each linked through <u>booking_id</u> .

CONCLUSION

In summary, this ERD outlines the core components of the International Airport Database system, encompassing airports, flights, airlines, passengers, bookings, boarding passes, luggage, and security screenings. The defined relationships explain how data flows across entities. Furthermore, primary and foreign key constraints enforce database integrity and logical organisation, serving as a solid architecture for managing airport operations efficiently.